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Title: Multi-Omics Data Integration for Non-small Cell Lung Cancer Metastasis

Abstract

One of the most common sites of metastasis from non-small cell lung cancer is bone. By integrating publicly available genomic and transcriptomic data, the genetic and molecular signatures that differentiate cells primed for bone colonization from other subtypes of the cancer will be identified. Likewise, a comparative analysis with the reciprocal pathway of osteosarcoma metastasis into the lung will help to identify any similarities or differences in these separate pathways. These computationally derived molecular signatures will further classify and clarify the specific pathways involved in lung cancer metastasis, as well as inform future research into its prevention or treatment.

Description

The question we are looking to answer is which mesenchymal cells in lung tumors are primed for bone metastasis. Using a variety of machine learning and computational methods, I will integrate several forms of omics data to isolate unique marker signals in lung cancer cells. This data will include genomic data, which refers to the DNA level of the gene, RNA sequencing data, which shows the transcription of genes in each cell, and proteomic data, which shows the proteins present in the cell (Lin et al. 2024). The different scales of data provide insight into several components of the pathways that cause the mesenchymal to epithelial transition that drives metastasis, as they cover transcription to

translation. The data can then be applied to train a machine learning algorithm, which can be used to recognize the differences between cells not primed for bone metastasis and cells that are likely to metastasize to bone (Preto et al 2024).

These prime differences classifying the cell subtypes are known as marker signals. These marker signals, unique gene expressions or molecular attributes, can be used to identify cells that are prone to metastasis into bone cancer, as they have statistically been determined as unique or enhanced in these types of cells. In addition, this data can be compared to other data describing the metastasis of osteosarcoma into the lungs, to see if there are any similarities between the two. This research can identify specific proteins or genes involved in the molecular pathways that drive bone colonization from lung cancer.

Previous Work

Cancer research has been focused on identifying tumor types or subgroups and quantifying distinct genetic markers for those types. In identifying the specific cancer occurring and the pathways that lead to tumor formation, treatment can be better targeted for each patient as the disease is more wholly understood. An important branch of cancer research is metastasis. Metastasis occurs when a tumor formed in one type of tissue migrates to another distant tissue and begins a new tumor colony (Lambert, Pattabiraman, & Weinberg, 2017). Metastasis occurs from a variety of different processes, including the activation of cell signaling pathways, dedifferentiation, abnormal gene

expression, and cell-cell interactions (Lambert, Pattabiraman, & Weinberg, 2017). By examining and analyzing data from each of these important regions, molecular or genetic signatures can be used to characterize different subtypes or behavior of tumor types.

Multi-omics data integration refers to the combination of multiple levels of cellular data, including genomics, transcriptomics, and proteomics (Baiao et al., 2025). A variety of machine learning, artificial intelligence, statistics, and dimensional reduction techniques are used when integrating data. For example, Preto et al. were able to distinguish several disease subtypes of inflammatory bowel diseases using the XGBoost classifier machine learning model, a type of decision tree (2024). Alternatively, Lin et al. used a Random Forest model to discover and select genetic disease factors for their study of ischemic strokes (2017). Another popular option for modelling is nonnegative matrix factorization, which can be seen being used by Abe and Shimamura to predict and reveal gene expression in a variety of tested cell types (2023). Though methods and research foci may change between these studies, they share many similarities. Models are trained using one dataset, often from publicly available data, learn what patterns in gene expression to recognize, and then are used to make predictions on another dataset based on its own transcriptional information. These methods are especially useful when taking in all the variables and information involved with cellular processes, and their versatility makes them an important tool for studying cancer and tumor formation.

One of the most common primary tumor sites is the lungs. Non-small-cell lung cancer (NSCLC) describes the majority of cancer in the lungs and is composed of adenocarcinoma and squamous cell carcinoma (Neimira et al., 2019). The second most common site of metastasis from NSCLC is bone (Wu et al, 2021). 30-40% of patients with NSCLC experience bone metastasis (Roato, 2014). In addition, the most common site of metastasis from osteosarcoma is into the lungs (Liu et al., 2025). While these pathways may not necessarily follow the same mechanisms, there are likely commonalities between the two cancers.

Significance

Lung cancer is a widely prevalent cancer, causing over 140,000 deaths in the US in 2019 (Niemira et al., 2019). Many patients with this type of cancer experience bone metastases, which greatly reduces survival expectancy (Roato, 2014). Non-small cell lung cancer is well studied, but the emerging field of multi-omics data integration brings a new approach towards classifying subtypes of the cancer for analysis. For example, in their studies, Niemira et al. looked only at RNA sequence data, but from this were able to determine 10 genes used to identify different subtypes of NSCLC and identify several pathways that may be the cause of their abnormalities (2019). While looking at every subgroup of lung cancer is important, my work will be focused specifically on bone metastases, narrowing the scope of the project and allowing us to focus more intensely on a few specific factors involved in the disease. Using multi-omics data will also allow for more precise identification of the pathways involved in the mesenchymal to epithelial transition, as there will be more data and evidence for

analysis and support. Understanding the pathways can help researchers decide which molecules to focus on in treatment and future research.

Proposed Methodology

Data will first be identified and collected from previous works, making sure that there is a certain degree of similarity between each sample, although there will be many differences that will have to be corrected for (Baião et al., 2025). While the data collection has not been completed yet, preliminary searches have identified a few key datasets for collection, including the transcriptomic data from Niemira et al. and proteomics data from Zhang et al.'s study on metastasis of lung adenocarcinoma. More data will be identified through searches of public databases such as the Gene Expression Omnibus (GEO) and the Cancer Genome Atlas. Then, a review of the scientific studies the data was taken from will verify their scientific relevance. Data will then undergo preprocessing and quality control in order to contribute the most accurate information towards the model. All analyses and computation will be done in RStudio, using published software and packages. Three different methods of data integration and analysis will then be tested. From the many available options for machine learning tools, I have selected XGBoost, unified nonnegative matrix factorization, and Random Forest as the three algorithms to be used in this study. The results from these methods can then be compared and may complement each other or reveal important insights into the computational methods themselves, as they each come with their own trade-offs.

After building the model, the results will then be analyzed. Unique biomarkers, such as a list of marker genes with high expression in migrating cells, can then be further researched to determine what pathways they are associated with. In their research, Gentles et al. combined data from 7 different sources of gene expression data to identify significant genes and create a molecular prognostic index, or a quantitative value used to predict survival risk for patients with lung cancer (2015). The biomarkers that I will be identifying will predict lung metastasis into bone. Once these genes are identified, additional research to clarify these results will include an analysis of the pathways contributing to these molecular markers. This process will include research into the literature as well, to further learn about and identify proteins or groups of genes important to the mesenchymal to epithelial process. To compliment this research, I will do an additional analysis of a dataset of osteosarcoma transcriptomics. Using the best method identified in my previous work, I will identify the molecular signatures of osteosarcoma metastasis into the lung. These results can then be qualitatively and quantitatively compared to my results from my previous analysis, to distinguish any outstanding similarities between the reciprocal pathway of metastasis. The comparison of this pathway in addition to the subtype identification of the lung cancer can then clearly describe the uniqueness of lung cancer metastasis into bone.

Proposed Schedule

Project Tasks	Summer 2026	Fall 2026	Winter 2027	Spring 2027
Data Analysis and Integration				
Data acquisition and batch correction				
Building computational tools				
Interpreting results				
Thesis Preparation				
Preparation of data figures and tables				
Writing methods and approach				
Literature review and writing				
Thesis Defense				

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